

Name: _____

In right $\triangle LMN$, the right angle is at N, $LN = 21$, $NM = 20$, $LM = 29$.

1. Write $\sin L$, $\cos L$, $\tan L$ as fractions and 4-place decimals.
2. Write $\sin M$, $\cos M$, $\tan M$ as fractions.
3. Which two ratios from 1-2 must be equal, and why?

Find x. Round to the nearest tenth.

4. A 52° angle has adjacent side 18 and opposite side x.
5. A 26° angle has hypotenuse 30 and opposite side x.
6. A 49° angle has hypotenuse 24 and adjacent side x.
7. A 38° angle has opposite side 16 and hypotenuse x.
8. A 63° angle has adjacent side 9 and hypotenuse x.
9. A 55° angle has opposite side 31 and adjacent side x.
10. You stand 320 ft from a 500-ft building. Find the angle of elevation to the top, to the nearest tenth.